Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is the main difference between the Skip-gram and Continuous Bag of Words (CBOW) models in the Word2Vec embedding approach? 1 / 1 point

- Vector dimensions
- The pretext task definition, i.e. the input-output relationship of text elements
- The core training algorithm
- Computational complexity

Nice work

In the Skip-gram model, the input is a target word and the output is context words, whereas in the CBOW model, the input consists of context words and the output is a target word.

2. Why are one-hot word vectors considered inadequate for capturing semantic similarity? 1 / 1 point

- They may contain noisy or missing features
- Even two words that are synonyms will share no features in common.
- They are difficult to update.
- They introduce random noise into word definitions.

Nice work

One-hot vectors assign a unique vector to each word with no shared features between synonyms, hence they cannot capture semantic similarity.

3. Why are word embeddings considered useful for NLP tasks like part-of-speech tagging and machine translation? 1 / 1 point

- Improving computational speed
- Understanding contextual semantics
- Enhancing syntactic structure

Nice work

Word embeddings provide insights into the semantic relationships and usage patterns of words in different contexts, which is essential for accurately tagging words with their corresponding parts of speech and translating words with their correct meaning.

4. What is the purpose of the softmax function in the output layer of the Skip-gram model? 1 / 1 point

- To reduce overfitting in the model
- To speed up the training process
- To normalize output probabilities

Nice work

The softmax function in the output layer of the Skip-gram model is used to convert raw output scores into probabilities by normalizing them, ensuring they sum to one and represent valid probabilities.

5. In a scenario where a new word embedding needs to be created whose vectors have similarities that reflect the global word co-occurrence statistics of those words across a large corpus, which embedding model should be used? 1 / 1 point

- Word2Vec
- GloVe
- One-hot encoding

Nice work

GloVe (Global Vectors) is designed to incorporate global co-occurrence information, making it ideal for learning word relationships in a large corpus by optimizing embeddings based on the frequency of word co-occurrences.

6. What is the primary purpose of the Skip-gram model in the Word2Vec algorithm? 1 / 1 point

- To predict context words from a target word
- To predict a target word from surrounding context words
- To cluster similar words together
- To categorize documents into topics

Nice work

The Skip-gram model is designed to take a target word as input and predict the likelihood of other words appearing in its context, thereby learning word embeddings based on the surrounding words.